

Finding Real Project Locations

Turning utility project lists into mapped, matched, verifiable data

What you're building

- A project table for **Utility A and Utility B**
- An **overlap table** — one row per pair of nearby projects owned by the two utilities

Utility A and B can be any two utilities whose service territories border or overlap. Our team used **Georgia Power (GPC)** and **Dominion Energy South Carolina (DESC)** as our working example

Three steps to find and match project locations

Each part below builds on the last: get coordinates → confirm they're right → calculate overlap.

Part 1 — Get coordinates and other data

Start here: the project list itself — names, in-service dates, already exists in each utility's own planning documents (IRPs, transmission expansion plans, SC RTP filings, or similar PDFs).

Folder with the two utility PDFs: [Project Listings](#)

- **Utility A project source:** [Dominion Energy](#)
- **Utility B project source:** [Georgia Power](#)

What's missing: coordinates. This part is about turning each project's name/description into a real-world location, plus filling in any other columns you're missing.

You're welcome to use any tool you find. Here's the approach we've used:

- Run queries against OpenStreetMap using [Overpass Turbo](#)
- Or search directly on [Nominatim](#) or [Open Infrastructure Map](#) to match a project's name/description to a real feature

The Overpass QL query below finds power substations by operator name within a bounding box. Swap in your utility's name and coordinates:

```
[out:json][timeout:25];
(
  nwr["power"="substation"]["operator"~"YOUR_UTILITY_NAME",i]
  (SOUTH_LAT,WEST_LON,NORTH_LAT,EAST_LON);
);
out body;
>;
out skel qt;
```

Paste this into the query box at [overpass-turbo.eu](#) and hit Run. Swap "power"="substation" for "power"="line" to pull transmission lines instead.

Tip — scale it up: You can run this query from the overpass-turbo.eu website, or from a Python backend hitting the Overpass API with the same query string. Either way: query for *all* of a utility's tagged infrastructure at once (every substation and line for that operator, not one at a time), export the result as **GeoJSON**, then filter that GeoJSON down to the features whose names/descriptions match your utility's

project list. That gives you one GeoJSON file per utility, already narrowed to the projects you actually care about, ready to plot or feed into Part 3.

Code Snippet: [Code Example](#)

Other sources worth checking for coordinates:

- [Overpass Turbo](#) — run OSM queries directly (what our snippet is built on)
- [OpenStreetMap](#) — browse and search the map directly for a project's location
- [Nominatim](#) — geocode a place name or address into coordinates directly
- [Open Infrastructure Map](#) — visual map of power infrastructure already plotted from OSM data, good for a quick sanity check

Part 2 — Confirm via public documentation

Public location data is limited, so double-check every match:

- Go back to the utility's PDF and re-read the project's description, zone, and nearby landmarks
- Compare that against the coordinate/feature you matched it to — a similarly-named substation or line in the wrong zone or county is a common false match
- **Match confirmed** → keep it
- **Can't confirm** → flag it as lower-confidence

Part 3 — Calculate distance and build the overlap table

Once both project tables have coordinates, calculate the distance between the center point of each Utility A project and the center point of each Utility B project (straight-line/haversine is fine — no need for driving distance). A project's center point is the midpoint between its two named sub-points; if only one point is located, that point is the center. Any pair under 25 miles apart counts as an overlap and gets one row in the overlap table. For each row, also record the time gap between the two projects' in-service dates, measured in days — distance is still the main signal for whether a pair overlaps at all, but the day count is worth a lighter look alongside it.

Project and Overlap Table: [Project|Overlap Table](#)

Starter file package for this project: [Starter File Package](#)